

V. Meta-Analysis of the Efficacy of Hormone Replacement Therapy in Treating and Preventing Osteoporosis in Postmenopausal Women

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A. Abstract

Objective: To review the effect of hormone replacement therapy (HRT) on bone density and fractures in postmenopausal women.

Data Source: We searched MEDLINE and EMBASE from 1966 to 1999, the Cochrane Controlled Register, citations of relevant articles, and proceedings of international meetings for eligible randomized controlled trials. We contacted osteoporosis investigators to identify additional studies, and primary authors for unpublished data.

Study Selection: We included 57 studies that randomized postmenopausal women to HRT or a control (placebo or calcium/vitamin D) and were of at least 1 yr in duration. Seven of these studies reported fractures.

Data Abstraction: For each study, three independent reviewers assessed the methodological quality and abstracted the data.

Data Synthesis: HRT showed a trend toward reduced incidence of vertebral fractures [relative risk (RR) 0.66, 95% confidence interval (CI) 0.41–1.07; 5 trials] and nonvertebral fractures (RR 0.87, 95% CI 0.71–1.08; 6 trials). HRT had a consistent effect on bone mineral density (BMD) at all sites. The difference between HRT and control in the percent change in bone density at 2 yr was 6.76 (5.83, 7.89; 21 trials) at the lumbar spine and 4.53 (3.68, 5.36; 14 trials) and 4.12 (3.45, 4.80; 9 trials) at the forearm and femoral neck, respectively.

Conclusions: HRT has a consistent, favorable and large effect on bone density at all sites. The data show a nonsignificant trend toward a reduced incidence in vertebral and nonvertebral fractures.

B. Introduction

ESTROGEN DEFICIENCY IS considered to be a major factor predisposing to osteoporosis. For years, experts have recommended HRT as a first-line therapy to prevent trabecular and cortical bone loss in postmenopausal women.

Abbreviations: BMD, Bone mineral density; CEE, conjugated equine estrogen; CHD, coronary heart disease; CI, confidence interval; DXA, dual x-ray absorptiometry; HERS, Heart and Estrogen/progestin Replacement Study; HRT, hormone replacement therapy; RCT, randomized controlled trial; RR, relative risk; QCT, quantitative computed tomography; WHI, Women's Health Initiative.

Randomized trials provide strong evidence that HRT prevents bone loss at both trabecular and cortical sites. However, the evidence for hip-fracture reduction comes primarily from case-control and cohort studies (1–5). The Heart and Estrogen/progestin Replacement Study (HERS) (6) evaluated the efficacy of HRT on the secondary prevention of heart disease; the investigators included fractures as a secondary endpoint. Although the women in this trial were not osteoporotic, there was no difference in hip fractures between those on estrogen compared with those on placebo. Recent recommendations have challenged the efficacy of HRT in the reduction of hip fractures, given the development of other therapeutic medications for osteoporosis.

In 1998, O'Connell *et al.* (7) published a systematic review of hormone replacement including trials from 1977 to 1995. This systematic review met major methodological criteria including a comprehensive search strategy and explicit inclusion and exclusion criteria. The authors included 37 studies in their review, but only one small trial provided data on vertebral fractures (8).

As part of this series of systematic reviews of osteoporosis therapy, the objective of this systematic review was to evaluate the magnitude of the efficacy of HRT on bone density and fractures.

C. Methods

1. Inclusion criteria. We included randomized controlled trials (RCTs) in postmenopausal women that compared an HRT intervention (estrogen used alone or estrogen opposed by a progestin) with a placebo. The HRT could be given in conjunction with a calcium and vitamin D supplement, provided the comparison group received the same supplements. Furthermore, the results had to be reported with a follow-up of at least 1 yr on one or more of the outcomes of interest: radiological evidence of fractures of the vertebra, wrist, or hip; or BMD measured in grams per centimeter or grams per centimeter squared by single-photon absorptiometry, dual-photon absorptiometry, dual x-ray absorptiometry (DXA), or quantitative computed tomography (QCT) at the lumbar spine, forearm, or femoral neck site. Both prevention and treatment trials were accepted. For this analysis, we applied stringent criteria when classifying trials as treatment, with treatment referring to studies that included women with prevalent fractures at baseline. This is in contrast with other

TABLE 1. HRT trial characteristics

Study (first author/ year/Ref.)	HRT preparation	Type of trial (duration in years)	No. of patients	Blinding	Outcomes measured
Abrahamsen, 1997 (12)	Trisequens	Prevention (2)	95	Open	BMD: Lumbar spine, forearm, femoral neck
Adami, 1989 (13)	Transdermal estradiol 50 μ g/d	Prevention (1.5)	34	Open	BMD: Forearm
Agnusdei, 1990 (14)	Premarin 0.625, 1-20, and cyclic provera	Treatment (1)	24	Open	BMD: Lumbar spine
Agnusdei, 1995 (15)	CEE 0.3 mg and cyclic provera	Prevention (1)	83	Blind	BMD: Distal forearm
Aloia, 1994 (16)	CEE 0.625 mg and cyclic provera	Prevention (2)	118	Open	BMD: Lumbar spine, femur
Alexandersen, 1999 (70)	Transdermal β estradiol 50 μ g/d Oral NETA, 1 mg/d	Prevention (2)	50	Blind	BMD: Lumbar spine, forearm, hip, femoral neck Fracture: Vertebral, nonvertebral
Cagnacci, 1991 (17)	Transdermal β estradiol 50 μ g/d	Prevention (2)	40	Unclear	BMD: Forearm
Castelo-Branco, 1992 (18)	CEE 0.625 mg transdermal 17 β estradiol	Prevention (1)	99	Open	BMD: Lumbar spine
Castelo-Branco, 1993 (19)	CEE 0.625 mg cyclic <i>vs.</i> continuous <i>vs.</i> cyclic transdermal estradiol 50 μ g	Prevention (1)	118	Unclear	BMD: Forearm
Cheng, 1993 (20)	Nylestriol 2 mg q 2 wk	Prevention (1)	283	Blind	BMD: Forearm
Christiansen, 1981 (21)	Trisequens forte	Prevention (2)	92	Blind	BMD: Forearm
Christiansen, 1981 (22)	Trisequens forte	Treatment (1)	264	Blind	BMD: Lumbar spine, forearm
Christiansen 1982 (23)	High, medium, and low dose sequential 17 β estradiol and 1 mg NETA	Prevention (1)	100	Blind	BMD: Forearm
Christiansen, 1990 (24)	2 mg of 17 β estradiol	Prevention (1)	40	Blind	BMD: Forearm
Civitelli, 1988 (25)	CEE 1.25 mg dial	Treatment (1)	21	Blind	BMD: Lumbar spine
Doren, 1995 (26)	2 mg continuous estradiol valerate	Prevention (2)	210	Open	BMD: Lumbar spine
Ettinger, 1992 (27)	0.5, 1.0, or 2.0 mg micronized estradiol	Prevention (1.5)	51	Blind	BMD: Lumbar spine
Finn Jensen, 1982 (29)	Trisequens	Treatment (1)	74	Blind	BMD: Forearm
Filipponi, 1995 (28)	Transcutaneous 17 β estradiol 50 μ g/d	Prevention (2)	124	Open	BMD: Lumbar spine
Gallagher, 1991 (30)	Premarin 0.3, 0.6 and provera 10 mg	Prevention (2)	81	Blind	BMD: Lumbar spine, forearm
Gambacciani, 1995 (31)	Oral ethinylestradiol 20 μ g/d	Prevention (1)	60	Open	BMD: Lumbar spine, forearm, total body
Gambacciani, 1997 (32)	0.3 mg CEE	Prevention (2)	80	Unclear	BMD: Lumbar spine
Genant, 1982 (33)	CEE 0.15, 0.30, 0.45, 0.60 daily	Prevention (2)	37	Blind	BMD: Lumbar spine
Genant, 1997 (34)	Esterified estrogens 0.3, 0.625, 1.25	Prevention (2)	406	Blind	BMD: Lumbar spine, femoral neck, total hip
Gonnelli, 1997 (36)	Transdermal 17 β estradiol 0.05 mg/d and cyclic provera	Treatment (2)	90	Open	BMD: Lumbar spine
Greenspan, 1998 (37)	CEE 0.625 mg	Prevention (2)	425	Blind	BMD: Lumbar spine, femoral neck, total body Fractures: Vertebral
Harris, 1991 (38)	0.3 or 0.625 mg of estrone sulfate d 1-25	Prevention (2)	120	Blind	BMD: Lumbar spine
Heikkinen, 1997 (39)	2 mg estradiol valerate	Prevention (2)	78	Blind	BMD: Femoral neck
Hesley, 1998 (40)	17 β estradiol patch 0.025, 0.05, and 0.01 mg	Prevention (2)	91	Blind	BMD: Lumbar spine, forearm
Hosking, 1998 (42)	CEE/Trisequens	Prevention (4)	1609	Open	BMD: Lumbar spine, forearm, hip Fractures: Nonvertebral
Horsman, 1977 (41)	Ethinyl estradiol 25 or 50 μ g	Prevention (2)	61	Open	BMD: Forearm
Hully (HERS), 1998 (6)	CEE 0.625/ provera	Prevention (4)	2764	Blind	Fractures: Vertebral, nonvertebral
Isaia, 1989 (43)	2 mg /d of estradiol valerate d 1-24 and 10 mg MPA d 14-24	Prevention (1)	57	Open	BMD: Lumbar spine
Komulainen, 1997 (44)	2 mg estradiol valerate	Prevention (5)	464	Open	Fractures: Nonvertebral
Lindsay, 1976 (45)	Mestranol 40 μ g/d and gestranol 200 mg cyclic	Prevention (5)	–	Open	BMD: Metacarpal
Lindsay, 1978 (46)	Mestranol 40 μ g/d and cyclic gestranol	Prevention (1)	30	Open	BMD: Metacarpal

TABLE 1. Continued

Study (first author/ year/Ref.)	HRT preparation	Type of trial (duration in years)	No. of patients	Blinding	Outcomes measured
Lindsay, 1984 (48)	CEE 0.15, 0.3, 0.625, 1.25	Prevention (2)	150	Blind	BMD: Metacarpal
Lindsay, 1990 (49)	CEE 0.625 mg/d and cyclic provera	Treatment (2)	50	Open	BMD: Lumbar spine, femoral neck
Lufkin, 1992 (8)	Transdermal estradiol 0.1	Treatment (1)	75	Blind	BMD: Lumbar spine Fractures: Vertebral
MacIntyre, 1988 (51)	Percutaneous estrogel—3 mg of 17 E ₂ /cyclicprogesterone 300 mg	Prevention (2)	70	Blind	BMD: Lumbar spine
Marslew, 1991 (52)	17β estradiol 1.5 mg/d	Prevention (2)	73	Blind	BMD: Lumbar spine, forearm
Marslew, 1992 (53)	Estradiol valerate 2 mg	Prevention (2)	62	Blind	BMD: Lumbar spine, forearm, calcaneus
Meschia, 1993 (54)	CEE 1.25 mg/d	Prevention (2)	95	Open	BMD: Lumbar spine
Mizunuma, 1997 (55)	CEE 0.3 and 0.625 with 2.5 mg MPA	Prevention (2)	52	Open	BMD: Lumbar spine, femoral neck
Munk-Jensen, 1988 (56)	Continuous <i>vs.</i> sequential treatment of 2 mg estradiol and 1 mg norethisterone	Prevention (1)	151	Blind	BMD: Lumbar spine, forearm
Palacios, 1995 (57)	1.5 mg/d of percutaneous 17β Estradiol or 0.625 mg of CEE	Prevention (1)	68	Open	BMD: Lumbar spine
Parez-Jaraiz, 1996 (58)	17β Estradiol 50 μg/d and 10 mg MPA for 10 d	Treatment (1)	104	Open	BMD: Total body
PEPI, 1996 (59)	CEE 0.625 mg, CEE 0.625/ cyclic MPA, CEE 0.625 mg/continuous MPA	Prevention (3)	—	Blind	BMD: Lumbar spine, hip
	CEE 0.625/ micronized 200 mg for d 1-12				
Recker, 1977 (60)	CEE 0.625 mg/d 1-21 and methyltestosterone 5 mg	Prevention (2)	60	Open	BMD: Forearm
Resch, 1990 (61)	Trisequens—estradiol 2 mg and norstherone acetate 1 mg daily	Treatment (1)	31	Blind	BMD: Forearm
Riis, 1987 (62)	3 mg of 17β estradiol d 1-24	Prevention (2)	53	Blind	BMD: Forearm, total body
Riis, 1987 (63)	Estradiol cream and oral progesterone 200 mg in 2nd yr	Prevention (2)	76	Blind	BMD: Lumbar spine, forearm
Riis, 1988 (64)	2 mg estradiol, 1 mg NETA	Prevention (2)	49	Blind	BMD: Lumbar spine, forearm, total body
Rosen, 1997 (65)	CEE 0.625 mg/cyclic MPA	Prevention (2)	236	Blind	BMD: Lumbar spine, femoral neck
Speroff, 1996 (66)	EE2 1.0, 2.5, 5.0, 10.0 μg and NA 1.0 mg	Prevention (2)	—	Blind	BMD: Lumbar spine
Wimalawansa, 1995 (68)	Estrogel 2.5 g/d and micronized progesterone	Prevention (2)	58	Unclear	BMD: Lumbar spine, femoral neck
Wimalawansa, 1998 (69)	Premarin 0.625 mg	Treatment (4)	72	Open	BMD: Lumbar spine, hip Fractures: Vertebral, nonvertebral

reviews, which considered studies that restricted their population to women whose mean bone density T-score was at least 2 sd values below peak bone mass as treatment studies; and when baseline bone density was unavailable, classified studies as treatment if the prevalence of vertebral fracture at baseline was greater than 20%; or, when fracture prevalence was unavailable, if the average age was above 62.

2. Search strategy. The Cochrane Collaborative approach for identifying RCTs as described by Dickersin *et al.* (9) and modified for the Cochrane Musculoskeletal Group guided our literature search. Text words included: bone density, postmenopausal, fractures, hormone replacement, estrogen, estrogen synthetic, progestational hormones. We searched MEDLINE and EMBASE databases from 1966 to 1999, the Cochrane Controlled Register, citations of relevant articles, and proceedings of international meetings. We contacted osteoporosis investi-

gators to identify additional studies, and primary authors for unpublished data. Two reviewers examined each title generated from the search and identified potentially eligible articles for which we obtained the abstracts (B.S., J.P.).

3. Methodological quality. A validated quality-assessment tool (10) was used to rate each eligible study. The scale evaluated the method and concealment of randomization, whether the patients and those measuring outcome were blind to the allocation of the intervention, and the extent and accounting of dropouts and withdrawals from the study. Interobserver agreement on the rating was measured using the κ coefficient of agreement (with greater than 0.60 taken as an indication of substantial strength of agreement).

4. Data collection. Two independent reviewers (B.S., J.P.) abstracted all information and data using standardized data

TABLE 2. RR of fracture after treatment with HRT

Fracture site	Study (first author/year/Ref.)	No. of patients	RR of fracture (95% CI)	Weighted RR of fracture (95% CI)
Vertebral	Lufkin, 1992 (8)	75	0.63 (0.28, 1.43)	0.66 (0.41, 1.07)
	Greenspan, 1998 (37)	425	0.70 (0.06, 7.55)	
	Wimalawansa, 1998 (69)	72	0.40 (0.09, 1.80)	
	Hulley, 1998 (6)	2763	0.69 (0.34, 1.38)	
	Alexandersen, 1999 (70)	50	2.78 (0.12, 65.09)	
Nonvertebral	Greenspan, 1998 (37)	425	0.70 (0.22, 2.22)	0.87 (0.71, 1.08)
	Komulainen, 1997 (44)	464	0.40 (0.16, 0.99)	
	Wimalawansa, 1998 (69)	72	1.00 (0.07, 14.79)	
	Hulley, 1998 (6)	2763	0.94 (0.75, 1.19)	
	Hosking, 1998 (42)	1609	0.98 (0.29, 3.34)	
	Alexandersen, 1999 (70)	50	0.31 (0.03, 2.76)	

abstraction forms, and a third reviewer (G.W.) verified the data. Abstraction included information on pertinent methodological aspects of the study design, characteristics of the participants, the specific estrogen and progestin preparations used, and the outcomes assessed. In particular, the mean percent change from baseline and SD values for bone density and the number of vertebral and nonvertebral fractures were abstracted.

5. Analysis. For the analysis of bone density, the weighted mean difference in bone density between HRT and control groups was calculated using the difference in the percent change from baseline in the treatment and placebo groups and their associated SD values. For the analysis of vertebral and nonvertebral fractures, the relative risk of fracture was calculated. Fleiss (11) has described the methods we used for pooling the results. For the pooled results, site-specific 95% CIs were calculated for bone density (lumbar spine, forearm, femoral neck) and fractures (vertebral, nonvertebral).

Data were initially pooled broadly across prevention and treatment trials, regardless of the estrogen dose, the type of preparation, or whether or not the estrogen was opposed. We conducted subgroup analysis for: 1) prevention *vs.* treatment; 2) opposed *vs.* unopposed estrogen; 3) estrogen dose (low-dose estrogen was equivalent to 0.3 mg Premarin, medium dose to 0.625 mg, and high dose 0.9 mg); and 4) type of estrogen preparation (transdermal, oral estradiol, and oral CEE/Premarin). When pooling results from the lumbar spine, we pooled dual-photon absorptiometry and DXA but not QCT. The precision error of QCT is not as good as DXA of the posterior/anterior spine. QCT provides a measure of volumetric density, and higher rates of bone change have been reported with QCT. We tested for homogeneity using a χ^2 test procedure (11) and assessed publication bias using funnel plots.

D. Results

1. Search results. Electronic and hand-searching resulted in a total of 310 published papers that addressed the relationship between HRT and BMD or fracture incidence. There were 62 RCTs fulfilling the inclusion criteria and comparing an estrogen preparation to a control either being placebo or calcium/vitamin D (6, 8, 12–71). Four trials were excluded: one trial used a progesterone agent only (67), and three trials used tibolone (35, 47, 50). A later report (71) of an included trial (42) was not used because the fracture data were not separately reported as vertebral and nonvertebral. Of the 57 RCTs included in this review, 47 were prevention and 10 were treatment trials.

Table 1 summarizes the characteristics of the 57 studies. Thirty-nine studies were published in 1990 or later, and of these studies, 17 were published after 1995. The estrogen and progestin preparations were diverse, as is evident from Table 1. The blinding status of the trials varied with 31 blinded, 22 not blinded, and 4 with unclear status. Although the follow-up ranged from 1 to 5 yr, most trials had a 1- or 2-yr study duration.

2. Fractures. We reviewed seven studies with estrogen that included data on fractures (Table 2). For all studies, with the exception of HERS, all participants including placebo/control subjects were either supplemented with calcium or encouraged to increase their dietary intakes. Four of the five studies provided similar results of a positive effect of HRT on vertebral fractures, but with wide CIs and statistically nonsignificant results (8, 37, 69, 70). The test for heterogeneity was not statistically significant. The resulting pooled estimate indicates a 34% reduction in RR in vertebral fractures (95% CI 0.41–1.07; Fig. 1).

Six studies measured the effect of HRT on nonvertebral fractures (6, 37, 42, 44, 69, 70). The study by Komulainen *et al.* (44) assigned 464 women to 1 of 4 groups: HRT, vitamin D alone, HRT and vitamin D, or placebo, and the pooled results of the two HRT groups compared with the combined vitamin D alone and placebo group showed a statistically significant relative risk reduction for nonvertebral fractures for HRT. For our analysis, only the HRT and placebo groups were used, with the result barely reaching significance. Three of the other studies had a RR reduction that was not statistically significant, and the other had a RR of 1.0. Although the individual study results were somewhat disparate, the test for heterogeneity was not statistically significant. The pooling of the study results indicates a reduction in the RR of 13% in nonvertebral fractures (95% CI 0.71–1.08; Fig. 2). Too few studies were available with vertebral and nonvertebral fracture outcomes to properly interpret the funnel plots for assessing publication bias.

3. Bone density. Combining the results for all prevention and treatment trials, for both opposed and unopposed estrogen, the pooled percent change in bone density was statistically significant in favor of HRT at all measurement sites (Fig. 3). After 1 yr, the percent change in bone density was 5.4% better at the lumbar spine with HRT and 3.0% and 2.5% better at the forearm and femoral neck, respectively. After 2 yr of treatment, the percent change in favor of HRT increased by

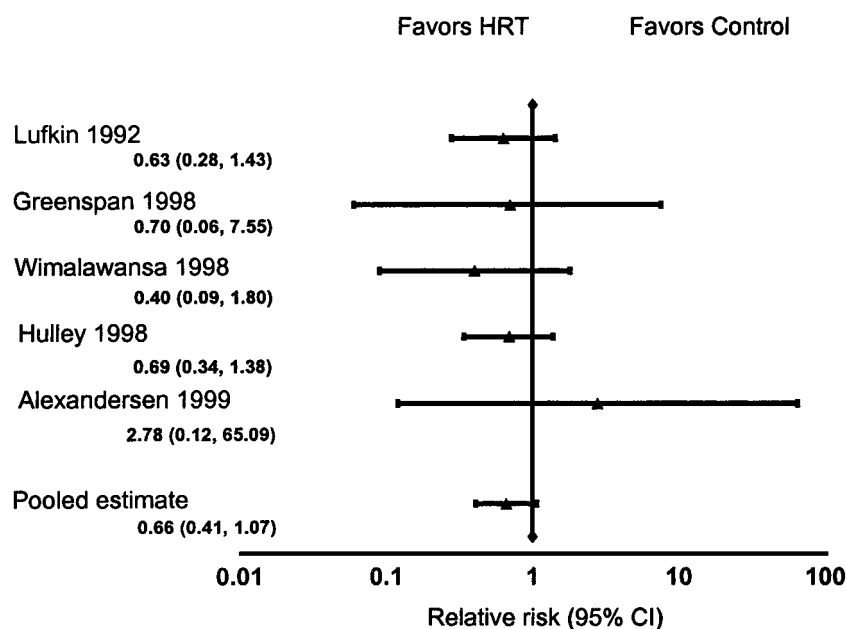


FIG. 1. RR of vertebral fracture after treatment with HRT.

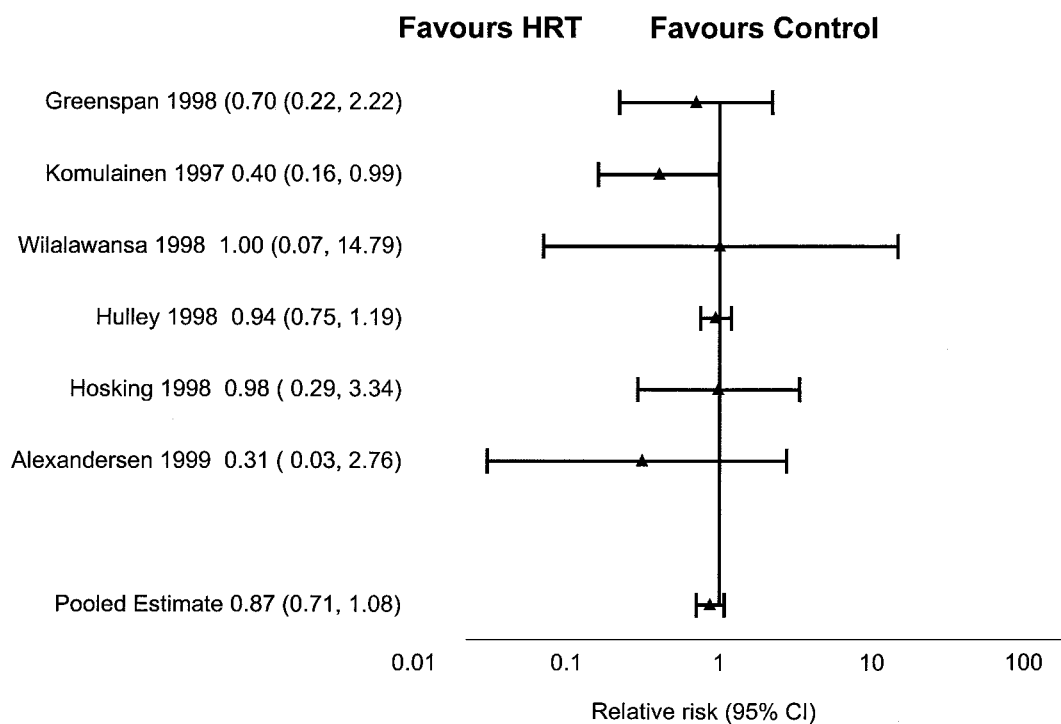


FIG. 2. RR of nonvertebral fracture after treatment with HRT.

about 1.5% at all sites with differences of 6.8%, 4.5%, and 4.1% for the lumbar spine, forearm, and femoral neck, respectively. There was no suggestion of publication bias from the funnel plots.

4. Subgroup analysis

a. Prevention vs. treatment. When we considered the prevention and treatment trials separately, the pooled percent change in bone density was still statistically significantly better with HRT after 1 yr at all measurement sites, and again the effect was

largest at the lumbar spine. For prevention after 1 yr, for both opposed and unopposed estrogen, the percent change in bone density was 4.9%, 3.0%, and 2.3% in favor of HRT at the lumbar spine, forearm, and femoral neck, respectively. The effect was greater when considering HRT for treatment with corresponding changes of 7.7%, 3.3%, and 3.5%, although the CIs were much wider than for prevention. At each of the sites, the difference between prevention and treatment was not statistically significant. The results were similar after 2 yr of therapy, with the pooled percent change in bone density significantly

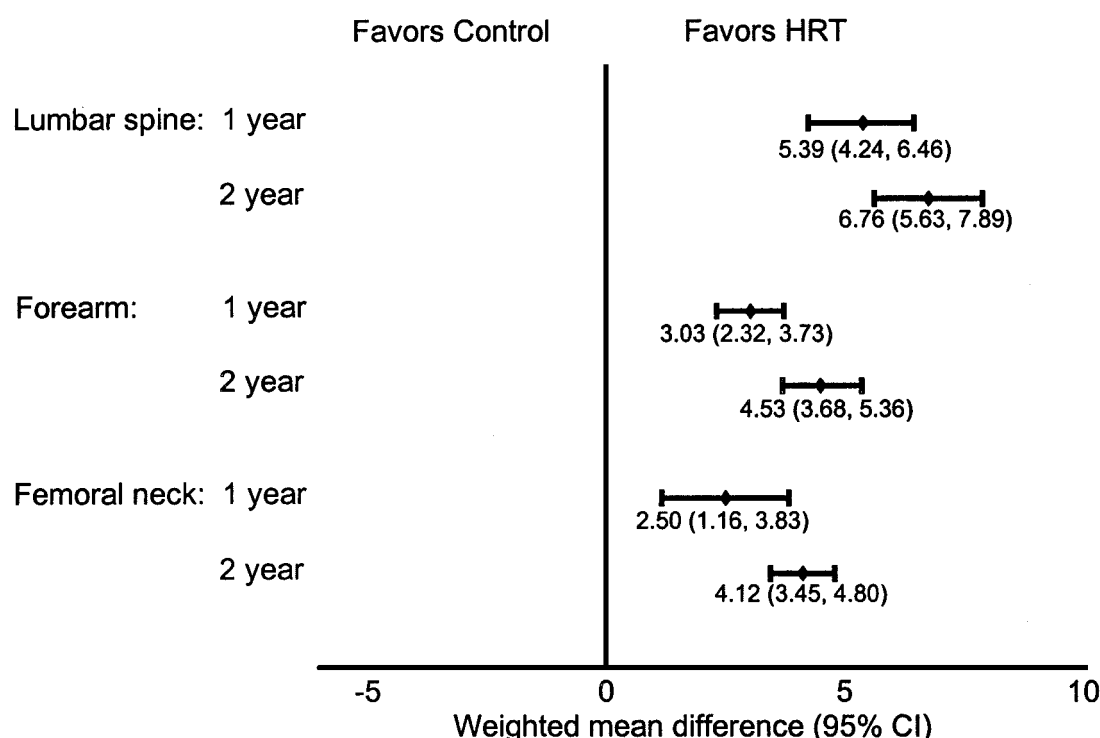


FIG. 3. Weighted mean difference in percent change in bone density after treatment with HRT.

TABLE 3. Weighted mean difference in percent change in bone density after treatment with HRT

Bone density site	Type of trial: (prevention/treatment)	Duration (years)	Weighted mean difference (95% CI) in the percent change in bone density		
			Opposed estrogen	Unopposed estrogen	Opposed/unopposed estrogen
Lumbar spine	Prevention	1	5.05 (4.36, 5.75)	3.86 (1.68, 6.03)	4.86 (3.70, 6.02)
		2	6.93 (5.63, 8.24)	6.75 (3.58, 9.91)	6.98 (5.53, 8.43)
	Treatment	1	7.25 (4.49, 10.11)	13.19 (8.01, 18.31)	7.70 (4.86, 10.54)
		2	5.65 (4.37, 6.94)	Not available	5.65 (4.37, 6.94)
Forearm	Prevention	1	3.03 (2.21, 3.86)	2.85 (1.08, 4.62)	3.01 (2.29, 3.74)
		2	4.53 (3.69, 5.36)	4.74 (3.33, 6.15)	4.53 (3.68, 5.36)
	Treatment	1	3.27 (0.35, 6.19)	Not available	3.27 (0.35, 6.19)
		2	Not available	Not available	Not available
Femoral neck	Prevention	1	2.50 (2.25, 2.77)	2.86 (1.90, 3.82)	2.25 (0.80, 3.69)
		2	4.07 (3.29, 4.84)	3.03 (1.88, 4.20)	4.07 (3.30, 4.84)
	Treatment	1	2.32 (−0.71, 5.35)	5.15 (1.27, 9.03)	3.46 (0.74, 6.19)
		2	4.70 (3.04, 6.36)	Not available	4.70 (3.04, 6.36)

better with HRT at all sites for both prevention and treatment (Table 3).

b. Opposed vs. unopposed estrogen. There were no significant differences when opposed and unopposed estrogen trials were compared at 1 yr. When we considered the opposed and unopposed estrogen trials separately, the pooled percent change in bone density for prevention trials was still statistically significantly better with HRT at all measurement sites and with the largest effect at the lumbar spine. For opposed estrogen, the percent change in bone density was 5.1%, 3.1%, and 2.5% in favor of HRT at the lumbar spine, forearm, and femoral neck, respectively, and for unopposed estrogen the corresponding percent changes were 3.9%, 2.9%, and 2.9%. The effects of HRT were similar but consistently larger, from 1% to 3%, after 2 yr of therapy. The only exception was for

unopposed estrogen at the femoral neck, which only showed a moderate increase (Table 3). For treatment trials with opposed estrogen, the pooled percent change in bone density was significantly better with HRT at the lumbar spine, with a 7.3% and 5.7% difference after 1 and 2 yr, respectively. Only a few treatment trials were available for assessing unopposed estrogen (Table 3).

c. Dose-response effect of estrogen. When we grouped the doses of estrogen according to low, medium, and high, we found a clear dose-response relationship at each site at 2 yr (Fig. 4). For low-dose estrogen (equivalent to 0.3 mg Premarin), the percent change in bone density was 3.9%, 3.1%, and 2.0% in favor of HRT at the lumbar spine, forearm, and femoral neck, respectively. The effect was greater for high-dose estrogen (equivalent to 0.9 mg Premarin), with corre-

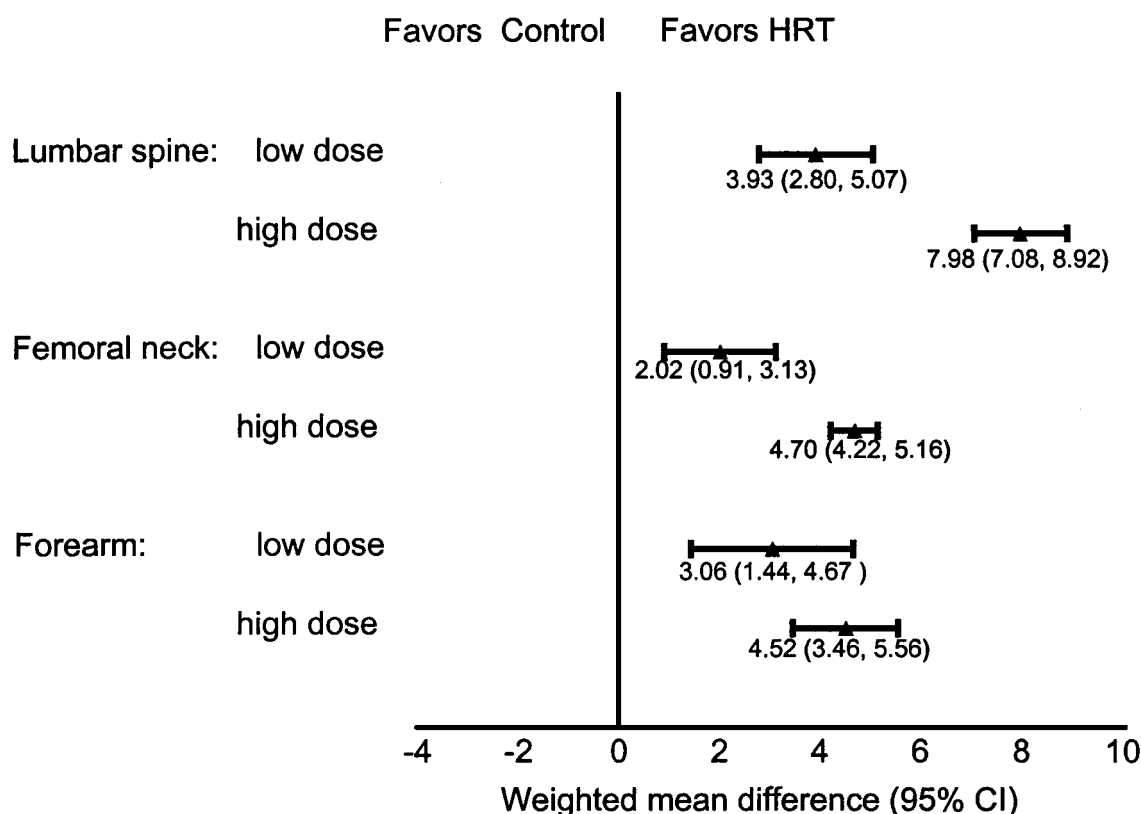


FIG. 4. Weighted mean difference in percent change in bone density after treatment with HRT: low dose *vs.* high dose after 2 yr of therapy.

sponding changes of 8.0%, 4.5%, and 4.7%. For the lumbar spine and femoral neck, there was a significant difference in the bone density at 2 yr, between the low and high dose of estrogen.

d. Type of estrogen preparation. When we grouped HRT by estrogen preparation, no differences in the bone density were found. For the lumbar spine, the difference between HRT and control in the percent change in bone density for the different types of estrogen preparations was 5.45 (3.31, 7.59) for transdermal, 5.36 (3.99, 6.75) for oral estradiol, and 5.62 (4.64, 6.60) for oral CEE/Premarin.

E. Discussion

This overview considered the impact of HRT on bone density and fractures. Our data suggest a strong impact of HRT on bone density at both trabecular and cortical sites after 1 and 2 yr of therapy. Furthermore, there appears to be a dose response in bone density with HRT. In particular, for the lumbar spine and femoral neck, there was a significant difference at 2 yr between low-dose estrogen, defined as equivalent to 0.3 mg of Premarin, and high dose, equivalent to 0.9 mg Premarin. We found no significant differences in the strong favorable impact of HRT on bone density at the various sites when investigators studied HRT for prevention or treatment purposes, when they used opposed or unopposed estrogen, or by the type of estrogen therapy (transdermal, oral estradiol or oral CEE/Premarin). Although the results strongly support a large positive effect of HRT on bone density, the duration of the RCTs were short, usually

1 or 2 yr, and longer-term evidence of the continued effect of HRT from such trials is needed.

Although evidence suggests that the risk of fracture is causally associated with BMD (72), there is a paucity of data from RCTs directly addressing the effect of HRT on fractures. Unlike the bisphosphonate studies in which the Food and Drug Administration demanded fracture data, such a demand was lacking for HRT, and so the motivation for fracture data was not the same. The limitations of the published fracture trials must be noted. Torgerson and Bell-Syer (73) reviewed HRT and the prevention of nonvertebral fractures and included 14 published papers on trials that reported fracture data. Of these trials, eight were excluded from this review: two controlled trials for concerns regarding the randomness of the allocation to the study groups (74, 75); two continuation studies of RCTs for concerns regarding selection bias that dramatically devalued the original RCT design (76, 77); one trial that focused on outcomes other than fractures, and the single fracture indicated is done only in the context of patients terminating the study and not as a comprehensive reporting on fractures (78); two trials that did not make a clear distinction of nonvertebral *vs.* vertebral fractures (79, 80); and one trial that reported fractures and not women with fractures (59). The study by Mosekilde (81) was not published in the timeline for this review. Even among the trials remaining and included in this review, limitations exist; in particular, with discontinuation rates ranging from 8% to 36%.

Torgerson and Bell-Syer (82) also reviewed HRT and the

prevention of vertebral fractures and included 10 published papers on trials that reported fracture data. Five of these studies were included in this review. Three trials were excluded: a continuation study of a RCT for concerns regarding selection bias that devalued the original RCT (83); a trial that did not make a clear distinction of nonvertebral *vs.* vertebral fractures (79); and a trial that reported fractures and not women with fractures (59). Two trials (81, 84) were outside the timeline of this review. Although not always clearly documented, all vertebral fractures were believed to be morphometric.

When we pooled the results from the existing randomized trials, we found a nonsignificant trend toward a reduction in both vertebral and nonvertebral fractures of 34% and 13%, respectively. If the study by Mosekilde (81), published outside the timeline of this review, was included, the overall results would not be substantially affected with nonsignificant reductions in the RR of vertebral and nonvertebral fractures of 21% and 16%, respectively. The Torgerson and Bell-Syer reviews yielded significant reductions in the RR of vertebral fractures of 33% (95% CI 0.45–0.98) and nonvertebral fractures of 27% (95% CI 0.56–0.94). Their reviews included a substantial amount of unpublished data and included trials that did not meet the methodological standards set for inclusion in our review. However, the trends in the reduction of the RRs of fractures are in a similar direction, and differences in statistical significance may be an issue of the precision of the reduction estimates due to the sample sizes involved.

Two studies currently in progress will provide needed fracture data: the Women's Health Initiative (WHI) Study and a British study, the Women's International Study of Long-Duration Estrogen after Menopause (WISDOM).

Although this review has focused on the effect of HRT on bone density and fractures, investigators have hypothesized that HRT reduces the risk of coronary heart disease (CHD) and other cardiovascular disease with increased breast cancer as a possible adverse outcome. Recommendations for HRT as preventive therapy against CHD is based on clinical studies demonstrating beneficial effects on low-density lipoprotein, high-density lipoprotein and lipoprotein (a) levels. Furthermore, there are extensive prospective and retrospective cohort and case-control data suggesting a reduction in the age-specific incidence of CHD in women. Virtually all epidemiological evidence accumulated and reviewed is from observational studies such as the Barrett-Connor and Grady (85) meta-analysis report. The HERS trial is the only RCT evaluating the effect of HRT on altering the risk of CHD in postmenopausal woman (6). Women with coronary disease were enrolled, and no differential effects of HRT on the occurrence of nonfatal myocardial infarction and CHD death, as well as secondary cardiovascular outcomes were observed. A second RCT of conjugated estrogen in postmenopausal women with CHD provides support for the HERS results (80). Also, the recent Women's Estrogen for Stroke Trial (WEST) (86) demonstrated that women assigned to estradiol had an increased risk of stroke in comparison to placebo during the initial 6 months. Given these results for secondary prevention in women with established coronary disease, the American Heart Association has included in its

current recommendation that "there is insufficient data to suggest that HRT would be initiated for the sole purpose of primary prevention of cardiovascular disease" (87).

Investigators have conducted several reviews of the effect of HRT on breast cancer. In particular, a study published by the Collaborative Group on Hormonal Factors in Breast Cancer (88) assembled and re-analyzed about 90% of the worldwide evidence on the relation between risk of breast cancer and use of HRT. Individual data on 52,705 women with breast cancer and 108,411 women without breast cancer from 51 studies in 21 countries were collected. They found that for current users of HRT or those who ceased use within 4 yr previously, the RR of having breast cancer diagnosed was 1.023 for each year of use (95% CI 1.011–1.036).

Historically, long-term HRT for postmenopausal women has attracted considerable attention in the research community and media for its potential in preventing death and disability in women from CHD and osteoporotic fractures. However, the potential role of HRT is currently controversial. The WHI will help clarify the role of HRT for the prevention of osteoporotic fractures, the reduction of CHD, and the increased breast cancer risk (89). The clinical trial component of the WHI is a RCT to evaluate three interventions: low-fat eating pattern, HRT, and calcium/vitamin D supplementation. In particular, HRT may reduce risk of CHD and other cardiovascular disease and secondarily to reduce the risk of hip and other fractures, with increased breast cancer as a possible adverse outcome. The planned completion date for the WHI is 2007.

Emerging information from studies such as the WHI will change the "landscape" of options for older women in the years to come (90), but the results will not be available for several years. For the treatment and prevention of osteoporosis in postmenopausal women, the current evidence indicates that HRT has a consistent effect on bone density at all sites, but has not established a beneficial effect of HRT on fractures, with evidence only showing a trend toward a reduction in fractures.

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